

Q: A camera captures an image that appears blurred due to improper sampling. Explain how sampling affects image quality and suggest a method to correct it.

Introduction

Sampling is one of the most important steps in digital image processing. It is the process of converting a continuous image into a digital image by dividing it into a grid of pixels. The quality of the digital image depends on how many pixels are used to represent the original image. If the sampling is not done properly, the image may lose important details and appear blurred or distorted.

How Sampling Affects Image Quality

Sampling determines the **spatial resolution** of an image. It specifies the number of pixels used to represent the image.

1. High Sampling Rate

- A high sampling rate means the image is captured using a large number of pixels.
- More image details are preserved.
- Edges appear sharp and clear.
- The image has better resolution and quality.

2. Low Sampling Rate (Improper Sampling)

- A low sampling rate means fewer pixels are used.
- Fine details cannot be represented accurately.
- The image appears blurred or pixelated.
- Important features may be lost.
- It may also produce **aliasing**, where edges appear jagged or false patterns appear in the image.

Why the Image Becomes Blurred

When the sampling frequency is lower than the amount of detail present in the scene, the camera fails to capture all the image information. As a result:

- Fine textures disappear.
- Object boundaries become unclear.
- Sharp edges become smooth.

- The overall image loses clarity and appears blurred.

Methods to Correct the Problem

The following methods can be used to improve image quality:

1. Increase the Sampling Rate (Best Solution)

- Capture the image using a camera with a higher resolution or more pixels.
- A higher sampling rate preserves more details and produces a sharper image.

2. Image Interpolation
If the image has already been captured, interpolation techniques can improve its appearance by estimating missing pixel values.

Common interpolation methods include:

- **Nearest Neighbor Interpolation** – Fast but may produce blocky images.
- **Bilinear Interpolation** – Produces smoother images by considering nearby pixels.
- **Bicubic Interpolation** – Uses more neighboring pixels and provides better image quality with smoother edges.

3. Super-Resolution Techniques

- Advanced image processing or AI-based algorithms combine information from one or more images to increase image resolution.
- These techniques recover finer details and improve sharpness.

4. Anti-Aliasing Filter

- Before sampling, a low-pass (anti-aliasing) filter is applied to remove high-frequency components.
- This reduces aliasing and preserves image quality.

Conclusion

Sampling directly affects the quality and resolution of a digital image. Improper or low sampling results in blurred images, loss of detail, and aliasing effects. The best way to prevent this problem is to use a higher sampling rate during image acquisition. If the image has already been captured, interpolation, super-resolution techniques, and image enhancement methods can improve its visual quality.